

1 & 2)

Exercise 2.1.1

1) $\int_0^3 \int_0^{\sqrt{9-y^2}} xy z^2 dx dy dz$ $x^2 + y^2 \leq 9$
 $x^2 \leq 9 - y^2$
 $x = \sqrt{9 - y^2}$

$= \int_0^3 \int_0^{\sqrt{9-y^2}} y z^2 (9 - y^2) dy dz$

$= \frac{1}{2} \int_0^3 \int_0^{\sqrt{9-y^2}} 9 y z^2 - y^3 z^2 dy dz$

$= \frac{1}{2} \int_0^3 9 y z^2 dy dz - \frac{1}{2} \int_0^3 y^3 z^2 dy dz$

$= \frac{1}{4} \int_0^3 81 z^2 dz - \frac{1}{8} \int_0^3 81 z^2 dz$

$= \frac{81}{8} \int_0^3 z^2 dz = \frac{27^2}{8} \approx 91,125$

2) i) $T = r(\cos \theta + \sin \theta \cos \varphi)$

$\nabla T = \begin{pmatrix} \cos \theta + \sin \theta \cos \varphi \\ \cos \varphi \cos \theta - \sin \theta \\ -\sin \varphi \end{pmatrix}$ $r = \sqrt{x_1^2 + x_2^2 + x_3^2}$

$\theta = \arccos\left(\frac{x_3}{\sqrt{x_1^2 + x_2^2 + x_3^2}}\right)$

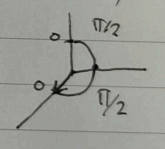
$\int_a^b (\nabla f) \cdot d\mathbf{I} = f(b) - f(a)$ $\varphi = \arctan\left(\frac{y}{x}\right)$

$T(2, 0, \varphi) = 2 \cdot (1 + 0) = 2$

$T(0, 0, \varphi) = 0$

$T(2, 0, \varphi) - T(0, 0, \varphi) = 2$

would be same for $\int_a^b (\nabla T) \cdot d\mathbf{I}$



3)

a)

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\boldsymbol{\alpha}$$

The left side means the divergence of vector $\mathbf{v}$ integrated over the volum of V . This gives "how much" flux the object has, as divergence is a measurment of "flux/volume". This could represent as a drain or a faucet showing the flow of the vector field.

The right side means the flux or the total flow of $\mathbf{v}$ through the surface S . This gives "how much" flux the object "sends out". Could be like the amount of water flowing through a dam.

Both sides give the total flow of the vector field within their areas, but through different dimensions.

b)

Saying the divergence is a larger at point means that the flow of the vector field is large right around that point. This does not mean the entire vector field has a large flow, just right around the point. Saying the flux through the surface is large means that the flow the vector field is large around the entire surface, making it more likely that the vector field is larger then if its just large around a point.

c)

Negative divergence corresponds to a drain, positive a faucet. The sum of the divergences in the volume equals the net flow through the boundary.

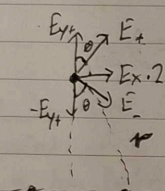
d)

3) a) $\nabla \cdot \mathbf{v} = 1$ $\int_V \nabla \cdot \mathbf{v} d\tau = 1$

$\int_0^1 \int_0^1 \mathbf{v} \cdot \hat{n} da$ | $x_2 x_3$ and $x_1 x_3$ - plane cancel

$\int_0^1 \int_0^1 (1-x) dx dy$ & $\int_0^1 \int_0^1 -(0-x) dx dy$

$1 - \frac{1}{2} = \frac{1}{2}$ $= \frac{1}{2}$ $\Rightarrow \frac{1}{2} + \frac{1}{2} = 1$

4) i)  $r^2 = z^2 + \frac{d^2}{2}$ $\sin \theta = \frac{d/2}{r}$

$2 \cdot E_x = 2 \cdot \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \sin \theta \hat{x} = E_{Total}$

$E_{Total} = \frac{1}{2\pi\epsilon_0} \frac{q d/2}{(z^2 + \frac{d^2}{2})^{3/2}} \hat{x}$

& The opposite $E_{Total} = \frac{1}{2\pi\epsilon_0} \frac{q d/2}{(z^2 + \frac{d^2}{2})^{3/2}} (-\hat{x})$

If $z \gg \frac{d}{2}$ then $\sin \theta = 0$ and $E_{Total} \Rightarrow 0$

5) i) $E(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\lambda(\mathbf{r}')}{r^2} \hat{r} d\tau'$

$r^2 = z^2 + a^2$ $\cos \theta = \frac{z}{r}$

$E = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{\lambda}{r^2} \cos \theta d\theta d\phi$

$= \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \frac{z}{(z^2 + a^2)^{3/2}} d\theta = \frac{\lambda}{2\epsilon_0} \frac{z}{(z^2 + a^2)^{3/2}} \hat{z}$

ii) $E = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{\lambda}{r^2} \cos \theta r dr d\theta$ | $r^2 = z^2 + r'^2$ $\cos \theta = \frac{z}{r}$

$= \frac{\lambda z}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \frac{1}{(z^2 + r'^2)^{3/2}} dr' d\theta$

$= \frac{\lambda z}{4\pi\epsilon_0} \int_0^{2\pi} \frac{1}{z - \sqrt{R^2 + z^2}} d\theta = \frac{\lambda}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{R^2 + z^2}} \right)$

4 & 5) ^^^^